XII LAPIS 2026 – Local Volume Cosmology

Student Project

Stellar Streams in Photometric Surveys: Building a Matched-Filter Pipeline

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Abstract

Students will build a main sequence turn off (MSTO) matched-filter to detect stellar streams in photometric data. Starting from the DELVE DR3 stellar catalog, they will construct a matched filter in colour magnitude space, produce filtered stellar density maps, an animation scanning over distance modulus, and recover known streams. Then they can create their own "Field of Streams" map showing structure in the Milky Way halo. Additionally, students can simulate their own stellar stream and inject it into real data to see what their search can and cannot find.

1 Goal

The goal of the project is to reproduce, with your own code, the photometric stream search of Shipp et al. (2018) on more recent DECam data.

The core project ends with filtered density maps in which you recover two or three known streams. From there you choose a direction:

- **Field of Streams.** Run your pipeline over a large area of the DELVE footprint and build a multi-distance colour map of halo substructure, in the spirit of Belokurov et al. (2006).
- **Simulate and inject.** Generate a synthetic stream with realistic photometric errors and completeness, inject it into the real catalog, and test whether your pipeline recovers it.

The questions you should be able to answer at the end:

- What does a stream look like in a colour magnitude diagram, and why do we use a MSTO matched filter isolate it?
- Which choices in the filter (age, metallicity, distance, colour width, magnitude limits) matter most, and how do you know?
- What else lives in a filtered halo map besides streams, and how do you tell them apart?

2 Dataset

2.1 Stellar catalog

You need positions, g and r magnitudes with errors (or you can try i, z), a star/galaxy classification, and quality flags for a region around each target stream. We will download **DELVE DR3 via LSDB**. The catalog is served publicly as a HATS catalog at `https://data.lsdb.io/hats/delve/delve_dr3_gold`. A box or cone cutout is a few lines of Python

with LSDB (https://github.com/astronomy-commons/lsdb). Keep only clean point sources (EXT_XGB = 0, FLAGS_GOLD = 0) and $16 < g < 24$. The project repository contains a notebook that downloads a region around the ATLAS-Aliqa Uma (AAU) stream.

Extinction. Interstellar dust dims and reddens stars, shifting them in the CMD by an amount that varies across the sky; an isochrone filter applied to uncorrected magnitudes will select different stars in different places. The _CORRECTED magnitudes have already been corrected using the Schlegel et al. (1998) dust map with the DES extinction coefficients, so use those columns.

2.2 Isochrones

Download PARSEC isochrones (Bressan et al., 2012) from the CMD web interface (http://stev.oapd.inaf.it/cgi-bin/cmd) in the DECam photometric system, or MIST isochrones (https://waps.cfa.harvard.edu/MIST/). You will need old ages (10–13.5 Gyr) and low metallicities ($Z = 10^{-4}$ to 10^{-3}). For AAU, see Li et al. (2021) for an appropriate isochrone.

2.3 Stream tracks and literature values

Use GALSTREAMS (Mateu, 2023) for the celestial tracks, coordinate frames, distances, and widths of known streams.

3 Software

There are two yaml files in the repo one to install all of the packages needed for a matched filter pipeline.

- numpy, scipy, pandas, matplotlib, astropy, healpy, skyproj.
- Data access: lsdb.
- Particle Spray simulations of streams: gala
- **streamobs** (https://github.com/LSSTDESC/streamobs), Track B only: the LSST DESC stellar stream injector. It generates stars from a parametric stream model, assigns photometry from an isochrone, and applies survey errors and completeness. It is not on PyPI: clone the repository, add it to your PYTHONPATH, and run bin/download_data.py. It depends on gala and on ugali (Bechtol et al., 2015), which must be installed from GitHub rather than PyPI (pip install git+https://github.com/DarkEnergySurvey/ugali.git).

4 Tasks

A matched-filter search for streams

1. **Build a stellar catalog.** Download a cutout around AAU (RA 5°–42°, Dec −42° to −14°). Apply star/galaxy separation, quality flags, and magnitude cuts (the _CORRECTED magnitudes are already extinction corrected). Plot the raw stellar density on the sky: the stream should *not* be visible.

2. **Rotate into stream coordinates.** Using the GALSTREAMS track, construct a great-circle frame (ϕ_1, ϕ_2) aligned with the stream and add these coordinates to your catalog.

3. **Find the stream in the colour–magnitude diagram.** Build an on-stream Hess diagram ($|\phi_2|$ small) and an off-stream one, normalize by area, and subtract. The stream population should appear as an overdensity tracing an isochrone.

4. **Choose an isochrone.** Download PARSEC isochrones and overlay them on the difference Hess diagram; choose an age, metallicity, and distance modulus.

5. **Construct the matched filter.** Select stars within a colour window around the isochrone whose half-width grows with the photometric error at each magnitude, $\Delta(g - r) = C + E\,\sigma(g)$, following Rockosi et al. (2002) and Shipp et al. (2018). Decide on bright and faint magnitude limits and justify them. *Extension:* replace the hard window with a weighted filter using the ratio of the stream and background Hess diagrams.

6. **Map and scan.** Bin the selected stars into a HEALPix map, smooth it, and display it. Repeat for a grid of distance moduli from 15.5 to 19 and show the stream appearing and disappearing. Compare the distance at which it is strongest with the literature value.

7. **Repeat for Phoenix and Tucana III.** These are at different distances, Phoenix is narrower, and Tucana III has a surviving progenitor. Note what you had to change. For each stream, compute a signal-to-noise ratio in the stream frame, $S/N = (N_{\rm on} - N_{\rm off}\,A_{\rm on}/A_{\rm off})/\sqrt{N_{\rm off}\,A_{\rm on}/A_{\rm off}}$, and compute the same statistic for an empty control field.

Option A: Build your own Field of Streams

A1. **Go wide.** Download a large region of DELVE (a few thousand deg^2; the whole southern Galactic cap if your connection and memory allow, in tiles). Keep the columns to a minimum.

A2. **Scan in distance.** Run your filter at distance moduli from 14 to 19 in steps of 0.25 and keep a HEALPix map for each.

A3. **Build the colour map.** Sum the maps in three distance ranges into red (far), green (middle), and blue (near) channels; stretch each channel; combine into an RGB image. Overlay GALSTREAMS tracks and the positions of globular clusters and dwarf galaxies.

A4. **Read the map.** Identify what you see: known streams, dwarf galaxies, globular clusters, the Sagittarius stream, the Magellanic Clouds and their periphery, survey depth variations, dust. Which features move between colour channels and which do not, and why? Are there overdensities you cannot match to a catalogued object?

Option B: simulate and inject your own stream

B1. **Install streamobs and generate a synthetic stream.** Configure a stream with a great-circle track through your control field, a Gaussian width of 0.2°–0.5°, a length of 10°–20°, a distance, and a total stellar mass or surface brightness. Use the DELVE error model and completeness. Inspect the output in the CMD and on the sky.

B2. **Inject and recover.** Append the synthetic stars to your real catalog and run your pipeline unchanged. Does the injected stream appear? At which distance modulus, and how does its S/N compare with the real streams?

B3. **Push until it breaks.** Make the stream fainter, more distant, wider, or give it a different age and metallicity from your filter isochrone. Find where your pipeline stops recovering it.

B4. **(If time allows) Detection efficiency.** Repeat B2 over a grid of surface brightness and distance with several random realizations each; define detection as S/N above a threshold set from the control field; plot efficiency versus surface brightness, one curve per distance, and mark where the real streams fall. This is the selection function needed to compare a stream census with simulations (Shipp et al., 2023).

Note. Pre-made AAU cutouts are available at https://drive.google.com/file/d/1wMQ4wCvaNaszgTu7Kd9
view?usp=drive_link

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